FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Main)-2023

TEST DATE: 27-03-2023

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

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Physics

PART – A

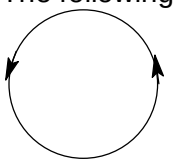
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. A solid cylinder rolling down an inclined plane (with angle of inclination θ) without slipping which of the following statement is incorrect?
(A) the friction force is dissipative
(B) the friction force will aid rotation
(C) the friction force will hinder translation
(D) the friction force will change if θ is changed

2. The following field line does not represent any of the two fields given in which option?



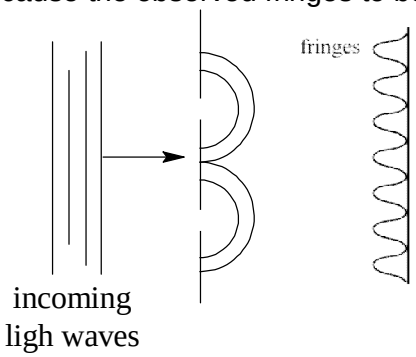
(A) Induced electric field and electrostatic field
(B) Magneto static field and induced electric field
(C) Gravitational field of a mass at rest and electrostatic field
(D) Magneto static field and Gravitational field of a mass at rest

3. A balloon filled with hydrogen has a volume 1 m^3 and its mass is 1 kg. Find the volume of a block of light material which it can just lift (in m^3): [density of block = 300 kg/m^3 and that of air = 2 kg/m^3]:
(A) $1/300$ (B) $1/250$
(C) $1/200$ (D) $1/100$

4. A large open tank has two holes in the wall. One is a square hole of side L at a depth y from the top and the other is a circular hole of radius R at a depth 4y from the top. When the tank is completely filled with water, the quantities of water flowing out per second from both holes are same. Then, R is equal to:
(A) $L/\sqrt{2\pi}$ (B) $2\pi L$
(C) L (D) $L/2\pi$

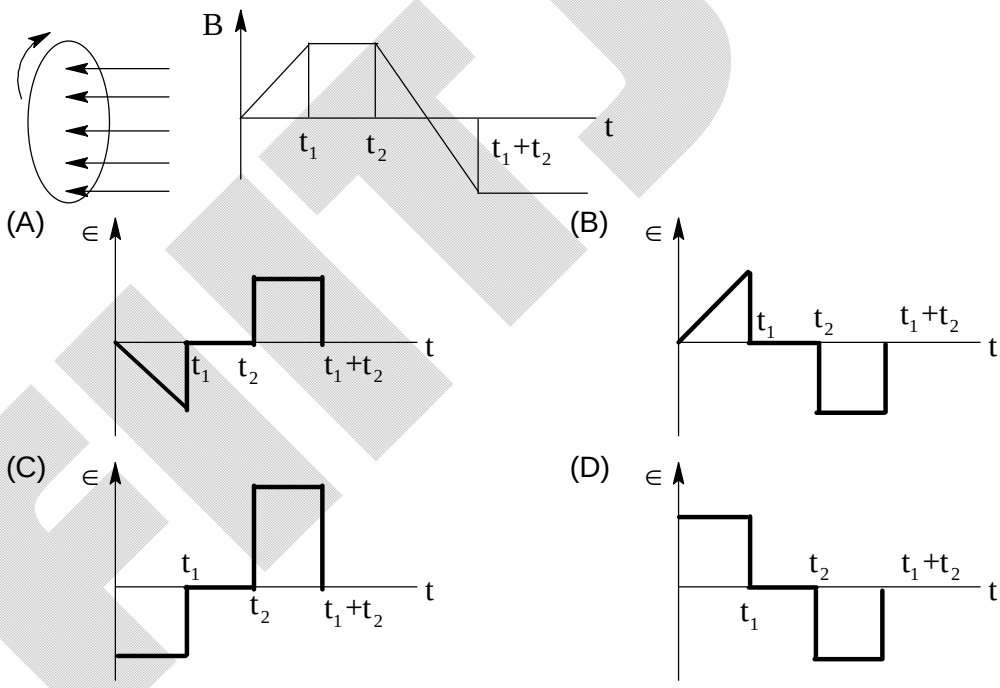
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5. Ina Young's double slit experiment, green light is incident on the two slits. The interference pattern is observed on a screen. Which of the following changes would cause the observed fringes to be more closely spaced?



(A) Reducing the separation between the slits
(B) Using blue light instead of green light
(C) Used red light instead of green light
(D) Moving the light source further away from the slits.

6. A wire loop is placed in a region of time varying magnetic field which is oriented orthogonally to the plane of the loop as shown in the figure. The graph shows the magnetic field variation as the function of time. Assume the positive emf is the one which drives a current in the clockwise direction and seen by the observer in the direction of B. Which of the following graphs best represents the induced emf as a function of time.

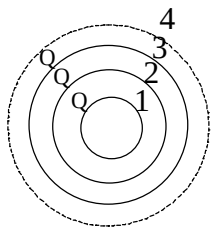


7. An object is kept at a distance of 16 cm from a thin lens and the image formed is real. If the object is kept at a distance of 6 cm from the same lens the image formed is virtual. If the size of the image formed are equal, the focal length of the lens will be:

(A) $19cm$ (B) $17cm$
(C) $21cm$ (D) $11cm$

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8. An infinite number of concentric rings carry a charge Q each alternately positive and negative. Their radii 1,2,4,8....meters in geometric progression as shown in the figure. The potential at the centre of the ring will be.



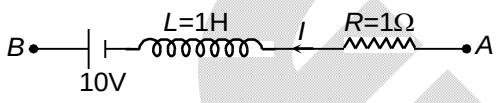
(A) Zero

(B) $\frac{Q}{12\pi\varepsilon_0}$

(C) $\frac{Q}{8\pi\varepsilon_0}$

(D) $\frac{Q}{6\pi\varepsilon_0}$

9. In the branch *AB* of a circuit, as shown in the figure, a current I = 5 A is flowing, where t is the time in second. At t =0, the value of $(V_A - V_B)$ is



(A) 15 V

(B) –5 V

(C) –15 V

(D) 5 V

10. A silver sphere (work function 4.6 eV) is suspended in a vacuum chamber by an insulating thread. Ultraviolet light of wavelength 0.2 μm strike on the sphere. The maximum electric potential of the sphere will be (hc = 12400 eVÅ)

(A) 4.6 V

(B) 6.2 V

(C) 1.6 V

(D) 3.2 V

11. The count rate observed from a radioactive source at t second was N_0 and at 4t second it was $\frac{N_0}{16}$. The count rate observed, at $\left(\frac{11}{2}\right)t$ second will be:

(A) $\frac{N_0}{128}$

(B) $\frac{N_0}{64}$

(C) $\frac{N_0}{32}$

(D) $\frac{N_0}{256}$

12. The rate of emission of radiation of a black body at 273°C is E, then the rate of emission of radiation of this body at 0°C will be:

(A) $\frac{E}{16}$

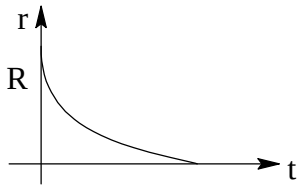
(B) $\frac{E}{4}$

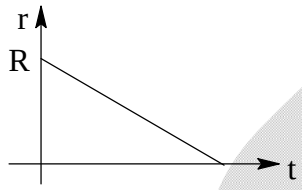
(C) $\frac{E}{8}$

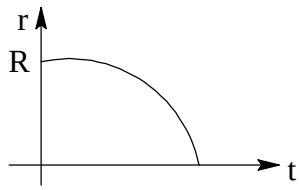
(D) 0

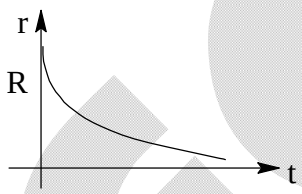
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13. A sphere of ice at 0°C having initial radius R is placed in an environment having ambient temperature > 0°C. The ice melts uniformly, such that shape remains spherical. After a time 't' the radius of the sphere has reduced to r. Assuming the rate of heat absorption is proportional to the surface area of the sphere at any moment, which graph best depicts r (t).

(A) 

(B) 

(C) 

(D) 

14. If angular velocity of a disc depends an angle rotated θ as $\omega = \theta^2 + 2\theta$, then its angular acceleration α at
$\theta = 1$ rad is :
(A) 8 rad/sec^2
(B) 10 rad/sec^2
(C) 12 rad/sec^2
(D) None of these

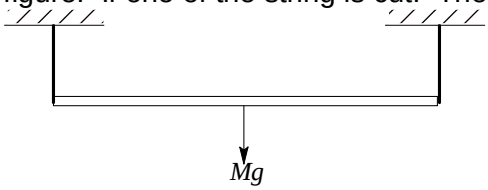
15. Two projectiles are thrown simultaneously in the same plane from the same point. If their velocities are v_1 and v_2 at angles θ_1 and θ_2 respectively from the horizontal. Then the trajectory of particle 1 with respect to particle 2 will be
(A) a parabola because both move in parabolic path.
(B) a straight line because both move in parabolic path
(C) a parabola because both move with same acceleration.
(D) a straight line because both move with same acceleration.

16. A simple pendulum is suspended from the ceiling of an empty box falling in air near earth surface. The total mass of system is *M*. The box experiences air resistance $\vec{R} = -k\vec{v}$ where *v* is the velocity of box and *k* is a positive constant. After some time it is found that period of oscillation of pendulum becomes double the value when it would have suspended from a point on earth. The velocity of box at that moment (take *g* in air same as on earth's surface)
(A) $\frac{Mg}{4k}$
(B) $\frac{3Mg}{4k}$
(C) $\frac{Mg}{2k}$
(D) $\frac{2Mg}{k}$

17. The average total energy density of the electromagnetic wave is, if the rms value of the electric field of the light coming from the sun is $500\,NC^{-1}$. :
(A) $4\times10^{-6}\ Jm^{-3}$
(B) $3.7\times10^{-19}\ Jm^{-3}$
(C) $1.35\times10^{-12}\ Jm^{-3}$
(D) None of these

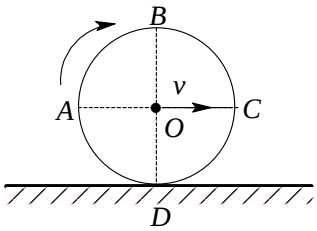
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18. A rod is supported horizontally by means of two strings of equal length as shown in figure. If one of the string is cut. Then tension in other string at the same instant will:



(A) remain unaffected
(B) increase
(C) decreases
(D) become equal to weight of the rod

19. A uniform circular ring rolls without slipping on a horizontal surface. At any instant, its position is as shown in the figure. Then mark the incorrect option:



(A) Section ABC has greater kinetic energy than section ADC
(B) Section BC has greater kinetic energy than section CD
(C) Section BC has the same kinetic energy as section DA
(D) None of these

20. A plane wave of sound travelling in air is incident upon a plane water surface. The angle of incidence is 60°. If velocity of sound in air and water are 330 m/s and 1400 m/s, then the wave undergoes:
(A) refraction only
(B) reflection only
(C) both reflection and refraction
(D) neither reflection nor refraction

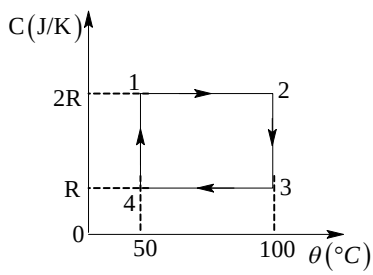
SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. If the ratio of escape velocity form the surface of two spherical planets of same mass density is 2. Find the ratio of radii of these planets.

22. Figure shows a cyclic process for one mole of a monoatomic gas where C is heat capacity and θ is temperature. The efficiency of cyclic process is $10\times K$ percent. Find the value of K :

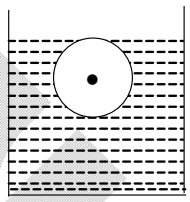


23. A ball of mass m makes head on elastic collision with a ball of mass nm which is initially at rest. Find the value of n for which fractional transfer of energy by the first ball is maximum.

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24. A rod having mass/length varying from its left end as $\lambda = \lambda_0(1+\alpha x)$ where $\lambda_0 = 1kg/m$ and $\alpha = 1m^{-1}$. The minimum moment of inertia of this rod about an axis perpendicular to rod is $\frac{211}{2^K \times 3^L}\text{kg}-\text{m}^2$. Find the value of (K+L)

25. A solid sphere of radius R just sink in water as shown in the figure. The force on lower hemisphere of sphere by the liquid is $\frac{k}{3}\pi R^3\rho g$, then find the value of 'k'.

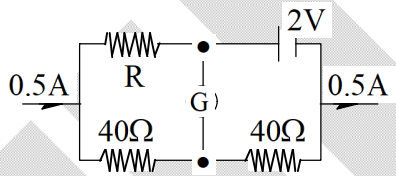


26. The work done in increasing the size of a soap film from $10\text{ cm} \times 6\text{ cm}$ to $10\text{ cm} \times 11\text{ cm}$ is $3\times10^{-4}\text{J}$. If surface tension (in Nm^{-1}) of the film is $\text{N x } 10^{-2}$. Then calculate N.

27. A string fixed at both ends is vibrating in the lowest mode of vibration for which a point at quarter of its length from one end is a point of maximum displacement. The frequency of vibration in this mode is 100 Hz the frequency emitted when it vibrates in the next mode such that this point is again a point of maximum displacement is $\eta \times 100\text{ Hz}$. Find the value of η.

28. An electric field is given by $\vec{E} = \left(y\hat{i} + x\hat{j}\right)\frac{N}{C}$. Find the work done (in J) in moving a 1C charge from $\vec{r}_A = \left(2\hat{i}+2\hat{j}\right)m$ to $\vec{r}_B = \left(4\hat{i}+\hat{j}\right)m$ ___ .

29. In the circuit shown in the figure, the internal resistance of the cell is negligible. For the value of R = 40/x Ω, no current flows through the galvanometer. What is x?



30. An object is kept at a distance of 16 cm from a thin leans and the image formed is real. If the object is kept at a distance of 6 cm from the same lens, the image formed is virtual. If the size of the images formed are equal, then find the focal length of the lens in cm:

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Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

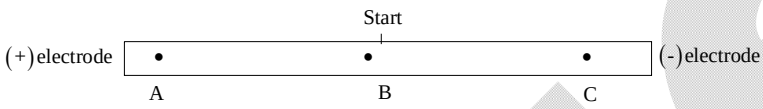
This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

31. Most soluble hydroxide in benzene is:

(A) $LiOH$ (B) $NaOH$

(C) KOH (D) $RbOH$

32. Referring to figure, identify the three spots obtained by electrophoresis at $p^H=7$ of a mixture of the proteins pepsin $(p^I=1.1)$ haemoglobin $(p^I=6.8)$ and Lysozyme $(p^I=11.0)$:



Which of the following options are correct about position of various proteins?

(A) Lysozyme will be present at spot A (B) pepsin will be present at spot B

(C) pepsin will be present at spot A (D) haemoglobin will be present at spot C

33. Which of the following is incorrect statement/reaction?

(A)



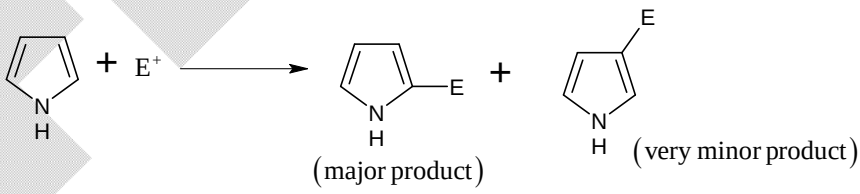
(B)



(C)

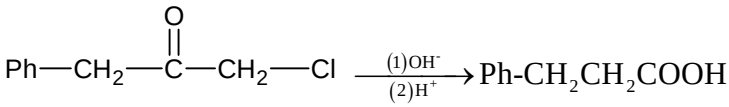


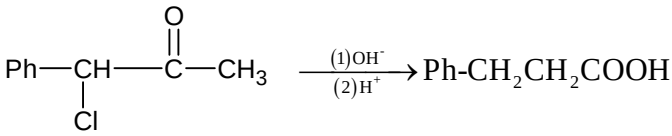
(D)

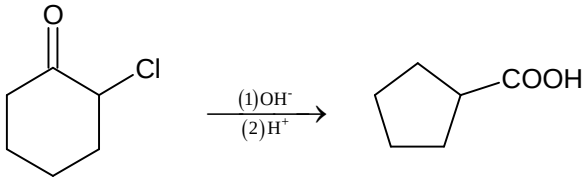


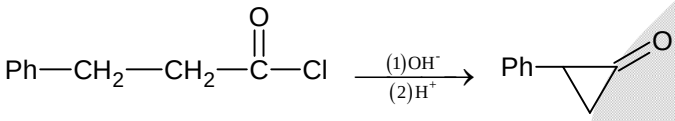
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34. Which of the following reaction is incorrect?

(A) 

(B) 

(C) 

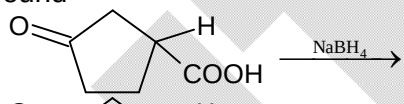
(D) 

35. What will be the ratio of volumes of $2M\ FeSO_4$ and $3M\ \ KMnO_4$ in acidic medium for oxidation of Fe^{2+} solution?

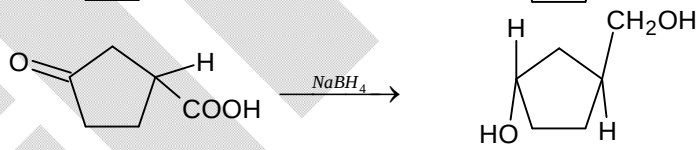
(A) $\frac{2}{9}$ (B) $\frac{15}{2}$

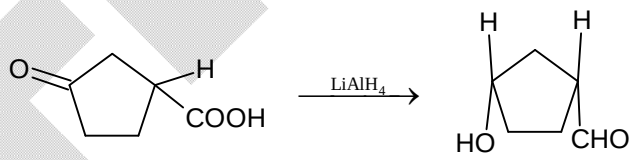
(C) $\frac{2}{15}$ (D) $\frac{9}{2}$

36. Which of the following reaction can be used for preparation of alcohol from carbonyl compound

(A) 

(B) 

(C) 

(D) 

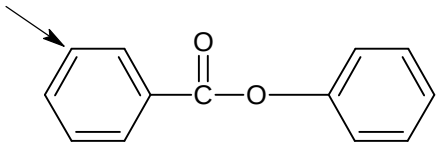
37. Which of the following molecules/ions are attracted by an externally applied magnetic field?

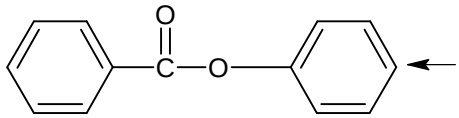
(A) O_2, NO, N_2^{2+}, B_2, C_2^{2+} (B) O_2^+, NO, N_2^+, C_2^{2+}, B_2

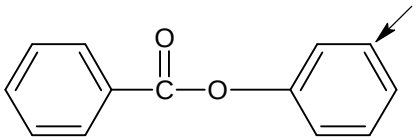
(C) N_2^{2-}, N_2^{2+}, C_2^{2-}, B_2^{2+}, O_2^{2+} (D) O_2^+, N_2^-, B_2^{2-}, NO, C_2^{2-}

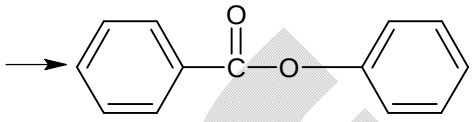
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38. When phenyl benzoate ($PhOCOPh$) undergoes electrophilic substitution, which of the following indicates the site for attack of electrophile as major product?

(A) 

(B) 

(C) 

(D) 

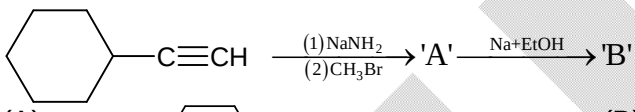
39. Which of the following statement about diffusion of gas is true?

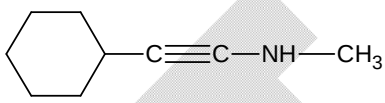
(A) Rate of diffusion of a gas will increase if an inert gas is added to gaseous mixture and allowed to diffuse through a pin hole

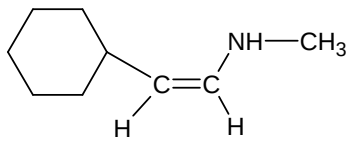
(B) Rate of effusion of O_2 is $\sqrt{2}$ times that of SO_2, for every composition of O_2 and SO_2 mixture

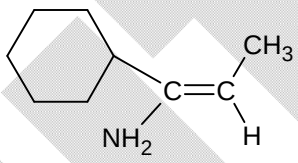
(C) Rate of effusion decrease on increasing the temperature

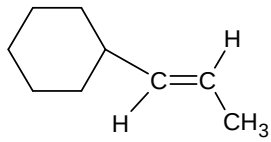
(D) All are true.

40. 

(A) 'A' is 

(B) 'B' is 

(C) 'A' is 

(D) 'B' is 

41. Which of the following redox reaction is balanced redox reaction?

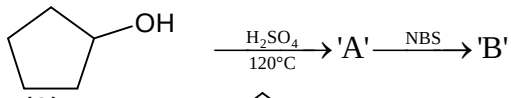
(A) $IO_3^- + SO_3^{2-} + 4H^+ \longrightarrow I^- + SO_4^{2-} + 2H_2O$

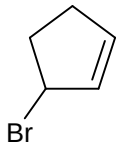
(B) $2BaO + O_2 \longrightarrow 2BaO_2$

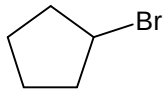
(C) $2I^- + 2IO_3^- \longrightarrow 2I_2 + 3O_2$

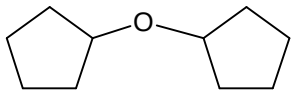
(D) $MnO_4^- + C_2O_4^{2-} + 8H^+ \longrightarrow M_n^{2+} + 2CO_2 + 4H_2O$

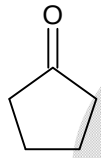
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42. 

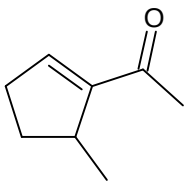
(A) 'B' is 

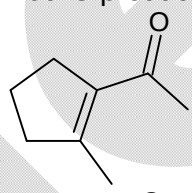
(B) 'B' is 

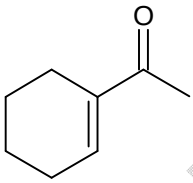
(C) 'A' is 

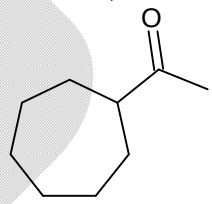
(D) 'A' is 

43. When Lithium di-(3-pentyl) cuprate reacts with ethyl bromide produces:
(A) heptane (B) triethyl methane
(C) 2-ethylpentane (D) 3-methyl hexane

44. 2, 7-Octanedione on treatment with aq. KOH and 100°C produces?
(A) 

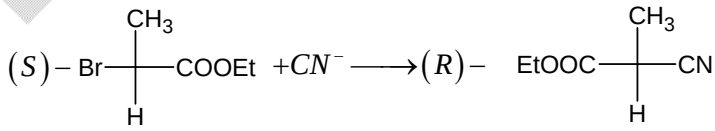
(B) 

(C) 

(D) 

45. Which of the following salts are soluble in NH_4OH?
(A) $CuSO_4$ (B) $AlCl_3$
(C) $FeCl_3$ (D) all of these

46. Zinc on reaction with NaOH gives a salt (A) along with a gas (X) and (A) on reaction with a gas H_2S gives white precipitate (W). Which of the following is correct?
(A) (A) is Na_4ZnO_3, (Y) is H_2S (B) (X) is H_2, (W) is $Zn(OH)_2$
(C) (A) is Na_2ZnO_2, (X) is O_2 (D) (W) is ZnS, (X) is H_2

47. Which of the following substitution reaction (SN^2) does not give product as shown is reactions?
(A) $(R)-CH_3CH(Br)CH_2CH_3 + MeO^- \longrightarrow (S)-CH_3CH(OMe)CH_2CH_3$
(B) $(S)-CH_3CH(Br)CH_2CH_3 + MeO^- \longrightarrow (R)-CH_3CH(OMe)CH_2CH_3$
(C) 

(D) cis-4-iodoethylcyclohexane+OH^- $\longrightarrow$ trans-4-ethylcyelohexanol

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48. The aqueous solution of which salt is coloured
(A) $Zn(NO_3)_2$ (B) $LiNO_3$
(C) $Co(NO_3)_2$ (D) Potash alum

49. If an electron has spin quantum number of $+\frac{1}{2}$ and magnetic quantum number of -1, it cannot be present in:
(A) $p_x - orbital$ (B) $d_{xy} - orbital$
(C) $p_y - orbital$ (D) $s - orbital$

50. Crystal field splitting energy (Δ_0) of complexes $\left[CrCl_6\right]^{3-}$, $\left[Cr(CN)_6\right]^{3-}$ and $\left[Cr(NH_3)_6\right]^{3+}$ are x, y and z cm^{-1} respectively. Then correct order Δ_0 values is:
(A) $x < y < z$ (B) $x < z < y$
(C) $x > y > z$ (D) $x > z > y$

SECTION – B

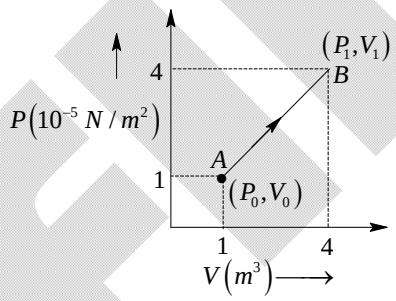
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. Silver sulphide dissolves in sodium cyanide solution to form the complex with coordination number?

52. How many five membered rings present in complex anion $\left[Co(EDTA)\right]^-$?

53. The work done in changing state from A to B in following system is -7.5×10^{-x} J, then the value of x is_____ .



54. $P_4 + HNO_3 \longrightarrow H_3PO_4 + NO_2 \uparrow + H_2O$
Product mixture is titrated with $0.1M$ $NaOH$, indicator shows colour change at second end point and $100ml$ of $NaOH$ is used for that. How much volume of $1M$ HNO_3 is used is above reaction? (in ml)

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55. Potassium crystallizes in BCC lattice with a unit cell length $a = 5.0\text{Å}$. What is the distance between next nearest neighbours? (in Å)

56. For the second order reaction; $2\text{A} \longrightarrow \text{B}$, time taken for the $[A]$ to fall to one-fourth value is how many times the time it takes for $[A]$ to fall to half of its initial value?

57. The equilibrium constant of the reaction is 10^{-x}. The value of 'x' will be:
$2\text{M}^{3+} + \text{N}^{+} \longrightarrow 2\text{M}^{2+} + \text{N}^{3+}$?
Given $\text{E}^{\circ}_{\text{N}^{+}/\text{N}} = 1.68\,\text{V}$ $\text{E}^{\circ}_{\text{N}^{3+}/\text{N}} = 1.50\,\text{V}$, $E^{\circ}_{M^{3+}/M^{2+}} = 0.75V$ and $\frac{2.303\ \text{RT}}{\text{F}} = .06$

58. The number of optically active complex compounds from the following:
(i) $\left[\text{Cr}(\text{ox})_3\right]^{3-}$
(ii) $\text{Cis}\left[\text{Co}(\text{en})_2\,\text{Cl}_2\right]^{+}$
(iii) $\text{Trans}\left[\text{Co}(\text{en})_2\,\text{Cl}_2\right]^{+}$
(iv) $\left[\text{Cr}(\text{en})_3\right]^{3+}$

59. The concentration of Ag^{+} is solution will just fail to give a precipitate of Ag_2CrO_4 with a solution in which $\left[\text{CrO}_4^{2-}\right] = 3\times10^{-4}\text{M}$ is $x\times10^{-4}\text{M}$. The value of 'x' will be:
$\left(\text{Given } \text{K}_{\text{sp}}\left(\text{Ag}_2\text{CrO}_4\right) = 1.2\times10^{-11}\right)$

60. A solution of liquids 'A' and 'B' having vapour pressure in pure state P_A^0 & P_B^0. The solution contains 30% mole of A which is in equilibrium with 60% mole of 'A' in vapour phase. If P_B^0 is 200 mm of Hg, then P_A^0 (in mm of Hg) is:

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Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. Which of the following function is surjective but not injective?

(A) $f:R\to R,\ f(x)=x^4+2x^3-x^2+1$ (B) $f:R\to R,\ f(x)=x^3+x+1$

(C) $f:R\to R^+,\ f(x)=\sqrt{1+x^2}$ (D) $f:R\to R,\ f(x)=x^3+2x^2-x+1$

62. If [x] denotes the integral part of x and in $(0, \pi)$, we define

$$f(x)=\begin{cases}\left[\dfrac{2(\sin x-\sin^n x)+|\sin x-\sin^n x|}{2(\sin x-\sin^n x)-|\sin x-\sin^n x|}\right], & x\neq\dfrac{\pi}{2}\\ 3, & x=\dfrac{\pi}{2}\end{cases}$$

Then for $n > 1$,

(A) $f(x)$ is continuous but not differentiable at $x=\dfrac{\pi}{2}$

(B) both continuous and differentiable at $x=\dfrac{\pi}{2}$

(C) neither continuous nor differentiable at $x=\dfrac{\pi}{2}$

(D) $\lim\limits_{x\to\frac{\pi}{2}} f(x)$ exists but $\lim\limits_{x\to\frac{\pi}{2}} f(x)\neq f\left(\dfrac{\pi}{2}\right)$

63. Let f be a function defined on R by $f(x)=[x]+\sqrt{x-[x]}$, then

(A) f is not continuous at every $x\in I$

(B) f is not continuous at every $x\in R\sim I$

(C) f is a continuous function

(D) none of these

64. Let $f(x)=\int\dfrac{x^2dx}{(1+x^2)(1+\sqrt{1+x^2})}$ and $f(0)=0$. Then $f(1)$ is

(A) $\log(1+\sqrt{2})$ (B) $\log(1+\sqrt{2})-\dfrac{\pi}{4}$

(C) $\log(1+\sqrt{2})+\dfrac{\pi}{4}$ (D) none of these

65. If statement $(p\to q)\to(q\to r)$ is false, then truth values of statement p,q,r respectively, can be-

(A) FTF (B) TTT

(C) FFF (D) FTT

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66. For $x > -1$, let $f(x) = \int_0^{\pi/4} \log_e(1 + x\tan z)\,dz$. Then value of $f\left(\frac{1}{2}\right) - f\left(\frac{1}{3}\right)$ equals

(A) $\frac{\pi}{4}\log_e\left(\frac{9}{8}\right)$ (B) $\frac{\pi}{8}\log_e\left(\frac{9}{8}\right)$

(C) $\frac{\pi}{9}\log_e\left(\frac{8}{9}\right)$ (D) $\frac{\pi}{8}\log_e\left(\frac{3}{2}\right)$

67. Area bounded by $y = \tan^{-1}x$, $y = \cot^{-1}x$ and y-axis is equal to

(A) $\ln\sqrt{2}$ sq. units (B) ln 4 sq. units

(C) ln 8 sq. units (D) ln 2 sq. units

68. If for $x < -1$, $\cos^{-1}\left(\frac{x^2-1}{x^2+1}\right) + \sin^{-1}\left(\frac{2x}{1+x^2}\right) - \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{\pi}{3}$, then x is equal to

(A) $-\sqrt{3}$ (B) $-\sqrt{2}$

(C) –2 (D) none of these

69. If $y = \frac{\cos 6x + 6\cos 4x + 15\cos 2x + 10}{\cos 5x + 5\cos 3x + 10\cos x}$, then $\frac{dy}{dx}$ is equal to

(A) $2\sin x + \cos x$ (B) $-2\sin x$

(C) $\cos 2x$ (D) $\sin 2x$

70. If in ΔABC, $\sin B\cos C = \frac{\sqrt{3}-1}{\sqrt{3}}$ and $\cos B\sin C = \frac{1}{\sqrt{3}}$, then the triangle is

(A) isosceles (B) equilateral

(C) right angled (D) right angled isosceles

71. The number of permutations of letters of word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears are

(A) 12246 (B) 169194

(C) 166680 (D) 181434

72. A is a 3 × 3 matrix with entries from the set {–1, 0, 1}. Then the probability that A is neither symmetric nor skew-symmetric is

(A) $\frac{3^9 - 3^6 - 3^3 + 1}{3^9}$ (B) $\frac{3^9 + 3^6 - 3^3 + 1}{3^9}$

(C) $\frac{3^9 - 3^6 + 3^3 + 1}{3^9}$ (D) none of these

73. If $(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then the value of $\sum_{k=0}^{3r} (-1)^k a_k\ {}^nC_{3r-k}$ (where $3r \le n$) equals

(A) $(-1)^r . {}^nC_r$ (B) ${}^nC_{3r}$

(C) $(-1)^{3r} . {}^nC_{3r}$ (D) ${}^nC_r$

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74. If $x \in R$ and $n \in I$, then the determinant $\Delta = \begin{vmatrix} \sin(n\pi) & \sin x - \cos x & \log\tan x \\ \cos x - \sin x & \cos\left[(2n+1)\frac{\pi}{2}\right] & \log\cot x \\ \log\cot x & \log\tan x & \tan(n\pi) \end{vmatrix}$ equals

(A) 0
(B) $\log\tan x - \log\cot x$
(C) $\tan\left(\frac{\pi}{4} - x\right)$
(D) none of these

75. If m is the root of the equation $(1-ab)x^2 - (a^2+b^2)x - (1+ab) = 0$ and m harmonic means are inserted between a and b, then the difference between the last and the first of the means equals

(A) $ab(a+b)$
(B) $a(b-a)$
(C) $ab(b-a)$
(D) $b-a$

76. If the equal sides AB and AC along positive x and y axes (each equal to a) of a right angled isosceles triangle ABC be produced to P and Q respectively so that $BP.\ CQ = AB^2$, then the line PQ always passes through a fixed point

(A) $(a, 0)$
(B) $(0, a)$
(C) (a, a)
(D) none of these

77. The equation of the circle, which is touched by the line $y = x$, has its centre on the positive direction of the x-axis and cut off a chord of length 2 units on the line $\sqrt{3}\ y = x$, is

(A) $x^2 + y^2 - 4x + 2 = 0$
(B) $x^2 + y^2 - 4x + 1 = 0$
(C) $x^2 + y^2 - 8x + 8 = 0$
(D) $x^2 + y^2 - 4y + 2 = 0$

78. If the tangents at two points (1, 2) and (3, 6) on a parabola intersect at the point (1, 6), then equation of directrix is

(A) $2x + y - 8 = 0$
(B) $x - 2y + 11 = 0$
(C) $x + 2y - 13 = 0$
(D) $2x - y + 4 = 0$

79. If a hyperbola whose foci are $S \equiv (2, 4)$ and $S' \equiv (8, -2)$ touches x-axis, then equation of hyperbola is

(A) $\frac{(x+y-6)^2}{10} - \frac{(x-y-4)^2}{8} = 1$
(B) $\frac{(x-y-4)^2}{10} - \frac{(x+y-6)^2}{8} = 1$
(C) $\frac{(x-y-4)^2}{20} - \frac{(x+y-6)^2}{16} = 1$
(D) $\frac{(x+y-6)^2}{20} - \frac{(x-y-4)^2}{16} = 1$

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80. The values of $x \in \left(0, \frac{\pi}{2}\right)$ satisfying the equation $\frac{\sqrt{3}-1}{\sin x}+\frac{\sqrt{3}+1}{\cos x}=4\sqrt{2}$ are

(A) $\frac{\pi}{36}, \frac{\pi}{12}$ (B) $\frac{\pi}{12}, \frac{11\pi}{36}$

(C) $\frac{12\pi}{36}, \frac{11\pi}{36}$ (D) $\frac{11\pi}{36}, \frac{\pi}{36}$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. The maximum value of area of rectangle *OABC*, where *O* is origin, *A* lies on positive *y*-axis and *B* lies on the curve $y=\frac{\ln x}{x^2}$, is $\frac{k}{e}$, then the value of 'k' is_____ .

82. Solution of $\sec^2 y \frac{dy}{dx}+x\tan y=x^3$ is $\tan y=x^2-k+ce^{-x^2/k}$, then find the value of 'k'

83. The tangents at z_1, z_2 on the circle $|z-z_0|=a$, meet at z_3. Then $\left(\frac{z_3-z_1}{z_1-z_0}\right)\left(\frac{z_0-z_2}{z_3-z_2}\right)$ is equal to_____ .

84. Number of value(s) of *x* satisfying the equation $3x^2-2x^3=\log_2\left(x^2+1\right)-\log_2 x$ is ___

85. If $A=\begin{bmatrix} 2 & 3 & 0 \\ 1 & \sin^2 x & -\cos^2 x \\ \cos^2 x & \cos^4 x & \sin^4 x \end{bmatrix}$ and $\mathrm{A\,(Adj\,A)}=\mathrm{K}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then the positive value of K is _____

86. Let S_n denote the sum to *n* terms of an arithmetic progression. If the common difference is equal to $S_n-kS_{n-1}+S_{n-2}$, then *k* =

87. If tangent and normal at point *P* (in first quadrant) to ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ intersect major axis at *T* and *N* respectively in such a way that ratio of area of ΔPTN and $\Delta PSS'$ is $\frac{91}{60}$, then area of $\Delta PSS'$ is (*S* and *S*′ are foci) $\mathrm{k}\sqrt{3}\,\mathrm{sq.}$ unit then the value of 'k' is ____.

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88. If $\sin A = x,\ \cos B = y,\ A + B + C = 0,$ then $x^2 + 2xy\sin C + y^2$ is equal to $k\cos^2 C$, then find the value of 'k' is _____.

89. The vector equation of the plane through the point $2\hat{i} - \hat{j} - 4\hat{k}$ and parallel to the plane $\vec{r}\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) - 7 = 0$ is $\vec{r}\left(4\hat{i} - 12\hat{j} - 3\hat{k}\right) = t$, then the value of 't' is _____

90. Standard Deviation of a data is 7, when each observation is decreased by 7 then standard deviation of new data is ____

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